Infinite Generation Two

— Native Pi Extension —

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1 Goal and acceptance criteria

The repository becomes a native Pi extension distributed from `https://github.com/FidesTherese/dsh-infinite-gen-2`. The extension MUST preserve the exact bytes of `prompts/infinite-gen-2.md`, expose the existing `infinite_gen2_profile` tool, append the bundled instructions without replacing Pi's base instructions, and display a session-local status indicator. The README MUST be English, contain accurate Pi installation instructions, and omit advertisements, donation requests, and promotional community links. The MIT notice and upstream attribution remain intact.

Acceptance requires automated tests, strict TypeScript checking, a real Pi loader/event-runner smoke test, and a package-content check. The requested handoff includes a reviewed commit pushed to the fork's existing `master` branch without rewriting history.

2 Original repository analysis

The inspected baseline is commit `3a701c3`. This is a small ES-module DeepSeek Harness plugin, not a standalone model client. Its execution path is `apply(ctx)` in `index.js`: read the bundled Markdown, register an ordered system-prompt section, register a profile tool, and optionally register an in-memory response-status projection. `client.js` consumes that projection through a host browser module loader and React, then inserts a badge into the conversation input dock. The Cordis patch declares the plugin bundle. Four shell/PowerShell scripts copy files and alter Harness profile configuration; the Windows installer also registers a desktop URI scheme. There are no application dependencies or automated tests declared at baseline.

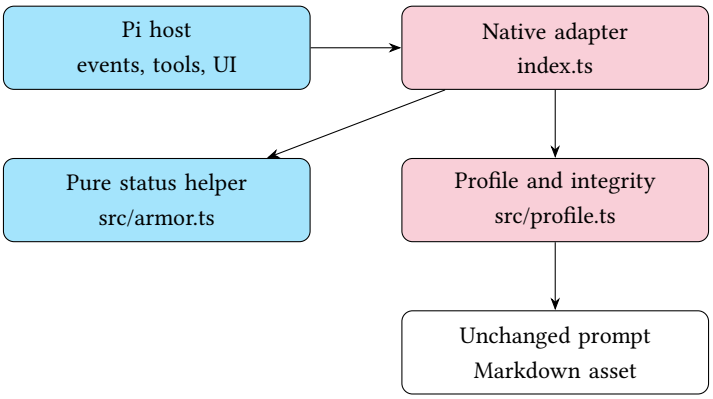
The host APIs, browser globals, and profile installers do not exist in Pi. The repository's reusable assets are the prompt, profile metadata, and the bilingual text-marker heuristic. The README's numeric strength comparisons are inherited calibration labels, not demonstrated benchmarks. The detector can report quoted phrases as refusals and cannot measure model compliance.

Recommended reading order is package metadata, the original entry point, the prompt asset, the projection scorer, and finally the browser/installer adapters. Git preserves these removed adapters at the baseline commit.

3 Chosen design and module boundaries

A direct TypeScript port is preferred to emulating Harness APIs or maintaining two host adapters: Pi already supplies prompt sections, tools, events, and status UI. `index.ts` owns Pi registration and ephemeral session state. `src/profile.ts` owns asset integrity and the read-only profile-tool contract. `src/armor.ts` owns pure text extraction, marker matching, and English status formatting. These files form one cohesive extension module; no service, database, browser, or provider integration is introduced.

The following nodes are new or replaced in this port. The prompt node is unchanged. Dependencies point from the adapter to its helpers, never back.



4 Interfaces, state, and critical flow

Pi discovers `index.ts` through the explicit package manifest. On load, the adapter verifies the prompt's SHA-256 and registers the tool, command, flag, and event handlers. Before each run, the adapter sets only its own `infinite_gen2` prompt section. Repeating the event replaces the same section rather than duplicating text; disabling removes only that section. It does not replace Pi's tools, project context, provider, or credentials.

The parameterless profile tool returns the complete unchanged prompt, its fingerprint, upstream metadata, prompt version 0.2.3, and extension version 0.3.0 as model-facing text, details, and structured JSON. The profile remains available while prompt injection is disabled.

The extension starts enabled. `--infinite-gen2-off` disables it at session startup. `/infinite-gen2 on|off|status` changes or reports the current session state; an empty argument means status. Commands never start an additional model turn. State resets on session startup/reload/replacement; a tree move clears the verdict. It is not durable configuration.

`agent_start` clears old results and marks the run busy. `message_end` inspects only assistant text, excluding thinking and tool blocks. Errors and cancellations, including automatic compaction failures, receive distinct outcomes rather than passing. Toggle commands preserve the busy state of an active run. `agent_settled`, not the low-level `agent_end`, ends the busy indicator after retries and queued work. Shutdown clears the status. The UI receives status only when available; headless model/tool behavior is identical and produces no

unsolicited stdout.

5 Constraints, risks, and non-goals

The target is Node.js 22.19 or newer and Pi 0.99.1 or a compatible newer release. TypeScript sources load directly through Pi; no compilation is required for installation. Pi and TypeBox are host-provided peer dependencies. Development versions are pinned in the lockfile.

A missing or changed prompt makes extension loading fail visibly; the port never substitutes another prompt. The frozen SHA-256 fingerprint is:

`02679eb726507d5a6bfe0a327c8469965440be7a7c5104cab57b8d1e529ec466`

Other extensions that force-replace the entire system prompt can override structured sections. Status scoring is an inherited text heuristic, not a security or effectiveness guarantee. The extension changes model instructions but does not alter provider safeguards, OS permissions, or tool approvals.

DeepSeek Harness compatibility, browser animations, URI-scheme installers, persistent toggles, automatic retries based on verdicts, telemetry, and prompt translation are out of scope. The obsolete Harness code is removed rather than advertised as supported. Implementation detail and the dependency-safe roadmap belong in `docs/DTD.tex`; module context belongs in `docs/modules/pi-extension.tex`.

6 Verification record

Verification on 2026-10-22: `npm test` exercises 31 focused SDK/unit cases; `npm run check` performs strict TypeScript checking; `npm run smoke` exercises three offline CLI/package-install cases; `npm pack --dry-run` confirms seven required distribution files. The package loads through the Pi 0.99.1 CLI and was also checked with the current Pi 1.4.0 CLI. Tests use isolated temporary Pi state and make no model requests. Live DeepSeek behavior is not an acceptance claim.